PassFrame tuition

Fraction Operations

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 2/4 + 1/4 = 3/4

2. B — 2/6

3. 4/6 − 1/6 = 3/6 (= 1/2)

4. 1/8 (multiply numerators: 1×1=1; multiply denominators: 4×2=8)

5. B — 5/8 (1/2 = 4/8, so 4/8 + 1/8 = 5/8)

6. 6/8 − 1/8 = 5/8

7. 3/4 cup (1/2 = 2/4, so 2/4 + 1/4 = 3/4)

8. A and C (1/2&1/4: 4 is a multiple of 2; 1/5&1/10: 10 is a multiple of 5. In B and D, neither denominator is a multiple of the other)

9. 3/9 + 1/9 = 4/9

10. A — 1/10

11. 5/6 − 2/6 = 3/6 (= 1/2)

12. 2/8 + 3/8 = 5/8

13. 4/8 + 3/8 = 7/8 litre

14. A — 5/10 (= 1/2) (1/5 = 2/10, so 7/10 − 2/10 = 5/10)

15. 8/12 + 1/12 = 9/12 (= 3/4)

Standard

1. 1/4 = 3/12, 1/6 = 2/12; 3/12 + 2/12 = 5/12

2. B — 13/20 (2/5 = 8/20, 1/4 = 5/20; 8/20 + 5/20 = 13/20)

3. 5/6 = 10/12, 1/4 = 3/12; 10/12 − 3/12 = 7/12

4. 2/3 × 1/4 = 2/12 (= 1/6)

5. B — 3/10 (3/4 × 2/5 = 6/20, which simplifies to 3/10)

6. 2/3 = 10/15, 1/5 = 3/15; 10/15 − 3/15 = 7/15

7. 2/5 = 6/15, 1/3 = 5/15; 6/15 + 5/15 = 11/15

8. A, B and C (3/6+2/6 = 5/6; 1/2+1/3 = 5/6 itself; 1/6×5 = 5/6; D gives 1 whole)

9. 3/8 + 2/8 = 5/8 (1/4 = 2/8)

10. A — 3/10

11. 7/8 − 4/8 = 3/8 (1/2 = 4/8)

12. 5/9 − 3/9 = 2/9 (1/3 = 3/9)

13. 3/5 = 12/20, 1/4 = 5/20; 12/20 + 5/20 = 17/20

14. C — 1/4 (1/3 × 3/4 = 3/12, which simplifies to 1/4)

15. 2/5 = 4/10; 4/10 + 3/10 − 1/10 = 6/10 (= 3/5)

Standard / Hard

1. 1/3 = 4/12, 1/4 = 3/12; 4/12 + 3/12 = 7/12

2. B — 1 7/12 (5/6 = 10/12, 3/4 = 9/12; 10/12 + 9/12 = 19/12 = 1 7/12)

3. 2/5 = 4/10; 9/10 − 4/10 = 5/10 (= 1/2)

4. 3/4 × 2/3 = 6/12 (= 1/2)

6. B — 11/18 (5/6 = 15/18, 2/9 = 4/18; 15/18 − 4/18 = 11/18)

7. 5/6 = 10/12; 10/12 + 1/12 = 11/12

8. A, B and D (3/4 × 1/3 = 3/12, which equals 1/4 and also 6/24; C, 1/3, is not equal)

9. 7/10 − 5/10 = 2/10; 1/5 = 2/10, so 2/10 + 2/10 = 4/10 (= 2/5)

11. 11/4 = 2 3/4; 2 3/4 − 1/4 = 2 2/4 (= 2 1/2)

12. 2/3 = 8/12; + 1/6 (2/12) = 10/12; − 1/4 (3/12) = 7/12

13. A and B (3/4+5/6 = 19/12 and 7/8+1/3 = 29/24, both greater than 1 whole; 1/2+1/3 = 5/6 and 2/5+1/4 = 13/20 are both less than 1)

14. 5/6 × 2/3 = 10/18 (= 5/9 m)

5. 3/4 × 2/3 = 6/12 (= 1/2 cup)

10. B — 2/5 (2/3 × 3/5 = 6/15, which simplifies to 2/5)

15. 9/10 − 6/10 = 3/10; 3/10 × 1/2 = 3/20

Homework — Answers

Easy / Standard

1. 2/6 + 1/6 = 3/6 (= 1/2)

2. B — 2/8

3. 6/8 − 1/8 = 5/8

4. 1/6 (multiply numerators: 1×1=1; denominators: 3×2=6)

5. B — 4/9 (1/3 = 3/9, so 3/9 + 1/9 = 4/9)

6. 5/8 − 2/8 = 3/8 (1/4 = 2/8)

7. 1/3 = 2/6, so 2/6 + 1/6 = 3/6 (= 1/2)

8. A and C (1/3 & 1/9: 9 is a multiple of 3; 1/5 & 1/10: 10 is a multiple of 5)

9. 3/6 + 1/6 = 4/6 (= 2/3)

10. B — 1/12

11. 7/8 − 2/8 = 5/8

12. 2/10 + 3/10 = 5/10 (= 1/2)

13. 4/8 + 1/8 = 5/8 m

14. B — 5/10 (= 1/2) (3/5 = 6/10, so 6/10 − 1/10 = 5/10)

15. 2/8 + 5/8 = 7/8

Standard

1. 1/4 = 3/12, 1/3 = 4/12; 3/12 + 4/12 = 7/12

2. B — 7/20 (3/4 = 15/20, 2/5 = 8/20; 15/20 − 8/20 = 7/20)

3. 5/9 − 3/9 = 2/9 (1/3 = 3/9)

4. 2/5 × 3/4 = 6/20 (= 3/10)

5. B — 1/3 (1/2 × 2/3 = 2/6, which simplifies to 1/3)

6. 7/10 − 5/10 = 2/10 (= 1/5)

7. 2/3 = 10/15, 1/5 = 3/15; 10/15 + 3/15 = 13/15

8. A, B and D (3/4 − 1/2 = 1/4; 2/8 = 1/4; 5/20 = 1/4; C, 3/6, equals 1/2, not 1/4)

9. 2/3 = 8/12; 8/12 + 3/12 = 11/12

10. B — 2/5 (4/5 × 1/2 = 4/10, which simplifies to 2/5)

11. 11/12 − 3/12 = 8/12 (= 2/3)

12. 3/8 = 9/24, 1/3 = 8/24; 9/24 + 8/24 = 17/24

13. 3/5 = 12/20, 1/4 = 5/20; 12/20 + 5/20 = 17/20

14. B — 1/4 (2/3 × 3/8 = 6/24, which simplifies to 1/4)

15. 5/6 = 10/12, 1/4 = 3/12, 1/3 = 4/12; 10/12 + 3/12 − 4/12 = 9/12 (= 3/4)

Standard / Advanced

1. Double 1/6 = 1/3; 1/3 + 1/3 = 2/3

2. Half of 2/3 = 1/3 (= 2/6); 2/6 − 1/6 = 1/6

3. A — 3/8 (1/4 + 1/2 = 3/4; half of 3/4 = 3/8)

4. Double 1/6 = 2/6; 5/6 − 2/6 = 3/6 (= 1/2)

5. A, B and D (half of 2/3 = 1/3; 2/6 = 1/3; 2/3 × 1/2 = 2/6 = 1/3; only C, 1/6, is not equal)

6. 5/6 = 10/12, 1/3 = 4/12, 1/4 = 3/12; 10 − 4 − 3 = 3, so 3/12 left (= 1/4)

7. 3/4 = 15/20, 1/5 = 4/20; 15/20 + 4/20 = 19/20, which is 1/20 short of the full target

8. A — 1/2 cup (2/3 × 3/4 = 6/12, which simplifies to 1/2)

9. 1/2 = 5/10, 1/5 = 2/10; used = 7/10; garden = 1 − 7/10 = 3/10

10. 3/4 × 1/3 = 3/12; 3/12 + 1/6 (2/12) = 5/12

11. A, B and D (1/4 = 2/8; + 1/2 (4/8) = 6/8; − 1/8 = 5/8, which is more than half, exactly 5/8, and less than 3/4 = 6/8; it is not exactly 1/2)

12. 1/3 = 4/12; two hoses add 2 × 1/4 = 1/2 = 6/12; total = 4/12 + 6/12 = 10/12 (= 5/6)

13. B — 17/20 (3/5 = 12/20, 1/4 = 5/20; 12/20 + 5/20 = 17/20)

14. 1/3 = 4/12, 1/4 = 3/12; 11/12 − 4/12 − 3/12 = 4/12 (= 1/3)

15. 5/6 − 1/4 = 10/12 − 3/12 = 7/12; 7/12 × 2/3 = 14/36 (= 7/18)